

Homework 1

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MA 726 Complex Geometry

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Problem 1 (1.1). *Let $U \subseteq \mathbb{C}^n$ be a connected open set. Reduce to the one-variable case to prove the following. Let $f : U \rightarrow \mathbb{C}$ be holomorphic. Suppose that $V \subseteq U$ is a non-empty open set with the property that f is identically zero on V . Then f is identically zero on U . Hence if $g \in \mathcal{O}_{\mathbb{C}^n}(U)$ is a second holomorphic function which agrees with f on V , then $f = g$ on all of U .*

Proof. Define the set

$$Z := \{z \in U \mid f \text{ vanishes on an open neighborhood of } z\}.$$

We will show $Z = U$. Since U is connected, it suffices to show Z is open and closed with respect to the subspace topology. Showing Z is open follows by definition: given $z \in Z$, let $K_z \subseteq U$ be an open neighborhood of z such that $f|_{K_z} = 0$. Then K_z is an open neighborhood of all $k \in K_z$ such that $f|_{K_z} = 0$, so $K_z \subseteq Z$. Thus

$$Z = \bigcup_{z \in Z} K_z.$$

Therefore we can realize Z as a union of open sets, meaning Z is open.

Recall that for all ξ in the closure $\overline{Z}$ (with respect to the subspace topology), any open neighborhood B of ξ intersects Z . Fix some $\xi \in \overline{Z}$. We want to construct B such that $f|_B = 0$.

Since polydisks form a basis for the topology on $\mathbb{C}^n$, there exists $\varepsilon_i > 0$, $1 \leq i \leq n$ such that $B := \Delta(\xi; \varepsilon) \subseteq U$. Since polydisks are connected, B is connected in the subspace topology. By assumption, $B \cap Z \neq \emptyset$. Throughout this proof, let $p = (p_1, \dots, p_n) \in B \cap Z$ such that $f(p) = 0$, and for each $1 \leq i \leq n$, write

$$K_i^p := \{z \in \mathbb{C} \mid (p_1, \dots, p_{i-1}, z, p_{i+1}, \dots, p_n) \in B\}, \quad g_i^p(z) := f(p_1, \dots, p_{i-1}, z, p_{i+1}, \dots, p_n).$$

Then $g_i^p : K_i^p \rightarrow \mathbb{C}$ is holomorphic in one variable since f is holomorphic. Write $\pi_i : \mathbb{C}^n \rightarrow \mathbb{C}$ for the i th projection $z \mapsto z_i$. Then $p_i \in K_i^p \cap \pi_i(Z \cap B)$, so $K_i^p \cap \pi_i(Z \cap B)$ is nonempty. We now show that K_i^p is open and connected.

Let $\phi_i : \mathbb{C} \rightarrow \mathbb{C}^n$ be the map $z \mapsto (p_1, \dots, p_{i-1}, z, p_{i+1}, \dots, p_n)$. Note that $K_i^p = \phi_i^{-1}(B)$, and that ϕ_i is clearly continuous since it is continuous in each variable. Since K_i^p is the preimage of an open set, it is open. Now consider the map $\pi_i : U \rightarrow \mathbb{C}$ given by $(z_1, \dots, z_n) \mapsto z_i$. This is well-known to be continuous, and clearly if $z \in K_i^p$ then $z \in \pi_i(B)$, so $K_i^p \subseteq \pi_i(B)$. Let $z = (z_1, \dots, z_n) \in B$ such that $\pi(z) = z_i$. We claim $z_i \in K_i^p$. It suffices to show that $(p_1, \dots, z_i, \dots, p_n) \in B$. Indeed, since $z \in B$ and $p \in B$, we have

$$|z_i - \xi_i| < \varepsilon_i, \quad \forall 1 \leq j \leq n, \quad |p_j - \xi_j| < \varepsilon_j.$$

Thus $\pi_i(B) = K_i^p$. Since images of connected sets are connected, K_i^p is connected.

Since K_i^p , Z , and B are open, and since π_i is an open map, we have that $K_i^p \cap \pi_i(Z \cap B)$ is open in $\mathbb{C}$. Since $g_i^p|_{\pi_i(Z \cap B)} = 0$, by the one-variable identity principle we have that $g_i^p = 0$. Now choose an arbitrary $z = (z_1, \dots, z_n) \in B$. By the above logic, $f(z_1, p_2, \dots, p_n) = g_1^p(p_1) = 0$. Thus f vanishes at $(z_1, p_2, \dots, p_n)$.

Now assume the inductive hypothesis that given $z \in B$ and $p \in Z$, we have $f(z_1, \dots, z_i, p_{i+1}, \dots, p_n) = 0$. It will be useful later to note that z and p need not be fixed. To show that $f(z_1, \dots, z_{i+1}, p_{i+2}, \dots, p_n) = 0$, it suffices to show that $\zeta := (z_1, \dots, z_i, p_{i+1}, \dots, p_n) \in Z$, since we could then apply the above logic to show the map $g_{i+1}^\zeta : K_{i+1}^\zeta \rightarrow \mathbb{C}$ is identically zero. Equivalently, it suffices to construct an open neighborhood $W \subseteq U$ of ζ such that $f|_W = 0$.

Since $p \in Z \cap B$, there exists a neighborhood $L \subseteq U$ such that $p \in L$ and $f|_L = 0$. Note that $L \subseteq Z$, and we can without loss of generality take L to be a polydisk $\Delta(p, \delta)$, since if it isn't, we can simply take a polydisk inside of it centered at p . We can write $\Delta(p, \delta) = \prod_{1 \leq j \leq n} \Delta(p_j, \delta_j)$ for each $\Delta(p_j, \delta_j) \subseteq \mathbb{C}$. Define

$$W := \left(\prod_{1 \leq j \leq i} \pi_j(B) \right) \times \left(\prod_{i+1 \leq k \leq n} \Delta(p_k; \delta_k) \right).$$

Clearly W is open, and moreover since $z_j \in B_j$ and $p_j \in \Delta(p_j; \delta_j)$ for all $1 \leq j \leq n$, we have $\zeta \in W$. Additionally, given $w \in W$, we have

- $(1 \leq j \leq i)$ $w_j \in \pi_j(B) = \Delta(\xi; \varepsilon)$ implies $|w_j - \xi_j| < \varepsilon_j$;
- $(i+1 \leq j \leq n)$ $w_j \in \Delta(p_j; \delta_j)$ implies that $|w_j - p_j| < \delta_j$. Since $\Delta(p; \delta) \subseteq \Delta(\xi; \varepsilon)$, we have $\Delta(p_j; \delta_j) \subseteq \Delta(\xi_j; \varepsilon_j)$, and thus $|w_j - \xi_j| < \varepsilon_j$.

Thus $w \in B$, and $W \subseteq B$. It suffices only to show $f|_W = 0$. To do this, we remark that since $z \in B$ was arbitrary, what the inductive hypothesis states is that for any $w \in B$ and any $p \in Z$, the map

$$(w_1, \dots, w_i) \mapsto f(w_1, \dots, w_i, p_{i+1}, \dots, p_n)$$

is identically zero. Let $w \in W$. Thus for $1 \leq j \leq i$, we have $w_j \in \pi_j(B)$, and otherwise we have $w_j \in \Delta(p_j; \delta_j) \subseteq L \subseteq Z$. Thus by the inductive hypothesis, we obtain that $f(w) = 0$, proving that $f|_W = 0$. Thus f vanishes on a neighborhood of ζ , and by induction we have $f|_B = 0$.

Since f vanishes on a neighborhood of ξ , it follows that $\xi \in Z$. Thus $\bar{Z} \subseteq Z$, and thus Z is closed. Since $f|_Z = 0$ and $Z = U$, we have $f = 0$.

The second statement follows by noting that if f, g are holomorphic on $U \subseteq \mathbb{C}^n$, then $f - g$ is holomorphic on U . ■

Problem 2 (1.3). (i) If $f : U \rightarrow \mathbb{C}$ is C^∞ on some open $U \subseteq \mathbb{C}^n$, then

$$\frac{\partial \bar{f}}{\partial \bar{z}} = \overline{\left(\frac{\partial f}{\partial z} \right)}, \quad \frac{\partial \bar{f}}{\partial z} = \overline{\left(\frac{\partial f}{\partial \bar{z}} \right)}.$$

(ii) Let $f : U \rightarrow \mathbb{C}$, $g : V \rightarrow \mathbb{C}$ be C^∞ on open $U, V \subseteq \mathbb{C}$ such that $f \circ g$ is defined. Then

$$\begin{aligned} \frac{\partial}{\partial z}(f \circ g)(z) &= \frac{\partial f}{\partial w} \frac{\partial g}{\partial z} + \frac{\partial f}{\partial \bar{w}} \frac{\partial \bar{g}}{\partial z} \\ \frac{\partial}{\partial \bar{z}}(f \circ g)(z) &= \frac{\partial f}{\partial \bar{w}} \frac{\partial \bar{g}}{\partial \bar{z}} + \frac{\partial f}{\partial w} \frac{\partial g}{\partial \bar{z}}. \end{aligned}$$

In particular, if f and g are analytic, then so is $f \circ g$ and the usual one-variable chain rule holds for ∂_z .

Proof. (i) I think $\partial_z f = \{\partial_{z_i} f\}_i$ so it suffices to show that $\overline{\partial_{z_i} f} = \partial_{\bar{z}_i} \bar{f}$. This is a direct computation. We write $f = u + iv$ for $u, v : \mathbb{R}^n \rightarrow \mathbb{R}^2$ and $z_i = x_i + iy_i$. Then recalling that $\partial_z = \frac{1}{2}(\partial_{x_i} - i\partial_{y_i})$ and similarly for $\partial_{\bar{z}}$,

$$\begin{aligned} \overline{\left(\frac{\partial f}{\partial z_i}\right)} &= \frac{1}{2} \overline{\left(\frac{\partial u}{\partial x_i} + i\frac{\partial v}{\partial x_i} - i\frac{\partial u}{\partial y_i} + \frac{\partial v}{\partial y_i}\right)} \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x_i} + \frac{\partial v}{\partial y_i} + i\left(\frac{\partial u}{\partial y_i} - \frac{\partial v}{\partial x_i}\right)\right) \\ \frac{\partial \bar{f}}{\partial \bar{z}_i} &= \frac{\partial}{\partial \bar{z}_i} (u - iv) \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x_i} - i\frac{\partial v}{\partial x_i} + i\frac{\partial u}{\partial y_i} + \frac{\partial v}{\partial y_i}\right) \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x_i} + \frac{\partial v}{\partial y_i} + i\left(\frac{\partial u}{\partial y_i} - \frac{\partial v}{\partial x_i}\right)\right). \end{aligned}$$

The calculations agree.

(ii) We can do this by direct computation using the chain rule for real-valued functions. Write $z = x + iy$ for the coordinates in V and $w = u + iv$ for the coordinates of U . For $f \circ g$ to exist, we must have $g : V \rightarrow U$. Write $g(x, y) = u(x, y) + iv(x, y)$ and $f(u, v) = \alpha(u, v) + i\beta(u, v)$. Then by definition, we have

$$(f \circ g)(x, y) = \alpha(g(x, y)) + i\beta(g(x, y)) = \alpha(u(x, y), v(x, y)) + i\beta(u(x, y), v(x, y)).$$

We then decompose the partial derivative as

$$\begin{aligned} \frac{\partial(f \circ g)}{\partial z} &= \frac{\partial}{\partial z} ((\alpha \circ g) + i(\beta \circ g)) \\ &= \frac{\partial(\alpha \circ g)}{\partial z} + i\frac{\partial(\beta \circ g)}{\partial z} \\ &= \frac{1}{2} \left(\frac{\partial(\alpha \circ g)}{\partial x} + i\frac{\partial(\beta \circ g)}{\partial x} - i\frac{\partial(\alpha \circ g)}{\partial y} - i^2\frac{\partial(\beta \circ g)}{\partial y} \right). \end{aligned}$$

Here we note that although g is complex-valued, we have written it as a function $\mathbb{R}^2 \rightarrow \mathbb{C}$, and since $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$, we have $\alpha \circ g, \beta \circ g : \mathbb{R}^2 \rightarrow \mathbb{R}$. Thus the usual chain rule applies.

$$\begin{aligned} \frac{\partial(\alpha \circ g)}{\partial x} &= \frac{\partial \alpha}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \alpha}{\partial v} \frac{\partial v}{\partial x} \\ \frac{\partial(\alpha \circ g)}{\partial y} &= \frac{\partial \alpha}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial \alpha}{\partial v} \frac{\partial v}{\partial y} \\ \frac{\partial(\beta \circ g)}{\partial x} &= \frac{\partial \beta}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \beta}{\partial v} \frac{\partial v}{\partial x} \\ \frac{\partial(\beta \circ g)}{\partial y} &= \frac{\partial \beta}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial \beta}{\partial v} \frac{\partial v}{\partial y}. \end{aligned}$$

This yields

$$\begin{aligned}
\frac{\partial(f \circ g)}{\partial z} &= \frac{1}{2} \left(\frac{\partial \alpha}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \alpha}{\partial v} \frac{\partial v}{\partial x} + i \frac{\partial \beta}{\partial u} \frac{\partial u}{\partial x} + i \frac{\partial \beta}{\partial v} \frac{\partial v}{\partial x} - i \left(\frac{\partial \alpha}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial \alpha}{\partial v} \frac{\partial v}{\partial y} + i \frac{\partial \beta}{\partial u} \frac{\partial u}{\partial y} + i \frac{\partial \beta}{\partial v} \frac{\partial v}{\partial y} \right) \right) \\
&= \frac{1}{2} \left(\frac{\partial \alpha}{\partial u} \left(\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \right) + \frac{\partial \alpha}{\partial v} \left(\frac{\partial v}{\partial x} - i \frac{\partial v}{\partial y} \right) + i \frac{\partial \beta}{\partial u} \left(\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \right) + i \frac{\partial \beta}{\partial v} \left(\frac{\partial v}{\partial x} - i \frac{\partial v}{\partial y} \right) \right) \\
&= \frac{\partial \alpha}{\partial u} \frac{\partial u}{\partial z} + \frac{\partial \alpha}{\partial v} \frac{\partial v}{\partial z} + i \frac{\partial \beta}{\partial u} \frac{\partial u}{\partial z} + i \frac{\partial \beta}{\partial v} \frac{\partial v}{\partial z} \\
&= \left(\frac{\partial \alpha}{\partial u} + i \frac{\partial \beta}{\partial u} \right) \frac{\partial u}{\partial z} + \left(\frac{\partial \alpha}{\partial v} + i \frac{\partial \beta}{\partial v} \right) \frac{\partial v}{\partial z} \\
&= \frac{\partial f}{\partial u} \frac{\partial u}{\partial z} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial z}.
\end{aligned}$$

Now note that

$$\begin{aligned}
\frac{\partial}{\partial w} + \frac{\partial}{\partial \bar{w}} &= \frac{1}{2} \left(\frac{\partial}{\partial u} - i \frac{\partial}{\partial v} + \frac{\partial}{\partial u} + i \frac{\partial}{\partial v} \right) = \frac{\partial}{\partial u}, \\
i \left(\frac{\partial}{\partial w} - \frac{\partial}{\partial \bar{w}} \right) &= \frac{1}{2} \left(i \frac{\partial}{\partial u} + \frac{\partial}{\partial v} - i \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) = \frac{\partial}{\partial v}.
\end{aligned}$$

This yields

$$\begin{aligned}
\frac{\partial(f \circ g)}{\partial z} &= \left(\frac{\partial f}{\partial w} + \frac{\partial f}{\partial \bar{w}} \right) \frac{\partial u}{\partial z} + i \left(\frac{\partial f}{\partial w} - \frac{\partial f}{\partial \bar{w}} \right) \frac{\partial v}{\partial z} \\
&= \frac{\partial f}{\partial w} \frac{\partial u}{\partial z} + i \frac{\partial f}{\partial w} \frac{\partial v}{\partial z} + \frac{\partial f}{\partial \bar{w}} \frac{\partial u}{\partial z} - i \frac{\partial f}{\partial \bar{w}} \frac{\partial v}{\partial z} \\
&= \frac{\partial f}{\partial w} \left(\frac{\partial u}{\partial z} + i \frac{\partial v}{\partial z} \right) + \frac{\partial f}{\partial \bar{w}} \left(\frac{\partial u}{\partial z} - i \frac{\partial v}{\partial z} \right) \\
&= \frac{\partial f}{\partial w} \frac{\partial g}{\partial z} + \frac{\partial f}{\partial \bar{w}} \frac{\partial \bar{g}}{\partial z}.
\end{aligned}$$

We now apply (i) to the first part of (2) to obtain the second part. Note that $\overline{(f \circ g)(z)} = \bar{f}(g(z))$, thus we can write $\overline{f \circ g} = \bar{f} \circ g$. We obtain

$$\begin{aligned}
\frac{\partial(f \circ g)}{\partial \bar{z}} &= \overline{\left(\frac{\partial(f \circ g)}{\partial z} \right)} \\
&= \overline{\left(\frac{\partial(\bar{f} \circ g)}{\partial z} \right)} \\
&= \overline{\frac{\partial \bar{f}}{\partial w} \frac{\partial g}{\partial z} + \frac{\partial \bar{f}}{\partial \bar{w}} \frac{\partial \bar{g}}{\partial z}} \\
&= \overline{\left(\frac{\partial \bar{f}}{\partial w} \right) \left(\frac{\partial g}{\partial z} \right) + \left(\frac{\partial \bar{f}}{\partial \bar{w}} \right) \left(\frac{\partial \bar{g}}{\partial z} \right)} \\
&= \frac{\partial f}{\partial \bar{w}} \frac{\partial \bar{g}}{\partial \bar{z}} + \frac{\partial f}{\partial w} \frac{\partial g}{\partial \bar{z}}.
\end{aligned}$$

When f, g are analytic, then they are holomorphic, so we have $\partial_{\bar{z}} g = \partial_{\bar{w}} f = 0$. We obtain

$$\frac{\partial f \circ g}{\partial z} = \frac{\partial f}{\partial w} \frac{\partial g}{\partial z}, \quad \frac{\partial f \circ g}{\partial \bar{z}} = 0.$$

■

Problem 3 (1.7). Working in $\mathbb{C}^2$ with variables z, w , find the Weierstrass polynomial for $\sin(w^2 - z^3)$.

Proof. Let $f(w, z) = \sin(w^2 - z^3)$ and let $p(z, w) = w^d + \alpha_{d-1}(z)w^{d-1} + \dots + \alpha_d(z)$ be the Weierstrass polynomial of $\sin(w^2 - z^3)$, which exists by the Weierstrass preparation theorem since $\sin(w^2 - z^3)$ is clearly holomorphic, $\sin 0 = 0$, and $\sin(w^2)$ is not identically zero.

Note that since $\alpha_i(0) = 0$ for all i , we have that $p(w, 0) = w^d$. In particular, $\partial_w^r p(0, 0) = 0$ for all $0 \leq r < d$. Note also that since u is a unit, it is nonzero at 0, so by the product rule

$$\frac{\partial f}{\partial w} = \frac{\partial u \cdot p}{\partial w} \Big|_{(0,0)} = \frac{\partial u}{\partial w} \Big|_{(0,0)} p(0, 0) + u(0, 0) \frac{\partial p}{\partial w} \Big|_{(0,0)} = u(0, 0) \frac{\partial p}{\partial w} \Big|_{(0,0)}.$$

In particular, $\partial_w^r f(0, 0) = 0$ if and only if $\partial_w^r p(0, 0) = 0$. Thus the degree d can be determined by taking partials of f . We compute:

$$\frac{\partial f}{\partial w} = 2w \cos(w^2 - z^3) \implies \frac{\partial f}{\partial w}(0, 0) = 0$$

$$\frac{\partial^2 f}{\partial w^2} = 2 \cos(w^2 - z^3) - 4w^2 \sin(w^2 - z^3) \implies \frac{\partial^2 f}{\partial w^2}(0, 0) = 2 \cos 0 = 2.$$

Thus the degree of the Weierstrass polynomial of f is 2.

We have

$$p(w, z) = w^2 + \alpha_1(z)w + \alpha_2(z).$$

The most natural thing to try here is to rewrite

$$\sin(w^2 - z^3) = (w^2 - z^3) \frac{\sin(w^2 - z^3)}{w^2 - z^3}.$$

Indeed, note that defining $\alpha_1(z) = 0$ and $\alpha_2(z) = -z^3$ yield the given polynomial, are both holomorphic, and vanish at 0. Thus $w^2 - z^3$ is the Weierstrass polynomial if the other term is a unit. This function has a removable singularity at 0, so we can extend it to a holomorphic function with

$$u(0, 0) = \lim_{(w,z) \rightarrow (0,0)} \frac{\sin(w^2 - z^3)}{w^2 - z^3} = 1.$$

Since u is nonvanishing at the origin and holomorphic, it is nonvanishing in a neighborhood of the origin, and thus is a unit for $\mathcal{O}_{\mathbb{C}^n, 0}$. Thus the Weierstrass polynomial of $\sin(w^2 - z^3)$ is $w^2 - z^3$. ■

Problem 4 (1.9). Denote by $\mathcal{A}_n$ the ring of germs at $0 \in \mathbb{C}^n$ of smooth functions. In contrast to $\mathcal{O}_n$, the ring $\mathcal{A}_n$ is not well-behaved algebraically. Specifically, prove that it is neither an integral domain nor Noetherian.

Proof. Recall that a function $\mathbb{C}^n \rightarrow \mathbb{C}$ is smooth if it is smooth as a function $\mathbb{R}^{2n} \rightarrow \mathbb{R}^2$. Thus it suffices to prove the statement for germs of smooth functions $\mathbb{R}^{2n} \rightarrow \mathbb{R}^2$. Define $f, g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} \exp\left(-\frac{1}{x_1^2}\right), & x_1 > 0 \\ 0, & x_1 \leq 0 \end{cases},$$

$$g(x) = \begin{cases} \exp\left(\frac{-1}{x_1^2}\right), & x_1 < 0 \\ 0, & x_1 \geq 0 \end{cases}.$$

Note that all partials of both functions clearly go to 0 as $x_1 \rightarrow 0$, so they are indeed smooth. However, since the sets on which they are nonzero are disjoint, $f \cdot g = 0$. Moreover, if we let $U = \{x \in \mathbb{R}^{2n} \mid x_1 > 0\}$ and $V = \{x \in \mathbb{R}^{2n} \mid x_1 < 0\}$, then 0 lies in the closures $0 \in \overline{U}, \overline{V}$. Thus any neighborhood of 0 intersects a region where $f, g \neq 0$, so neither are zero when passing to germs. Thus they are indeed zero divisors, and $\mathcal{A}_n$ is not an integral domain.

Now we show $\mathcal{A}_n$ does not satisfy the ascending chain condition. For each integer $k \geq 1$, define

$$S_k := \left\{ (re^{i\theta}, 0, \dots, 0) \in \mathbb{C}^n \mid 0 < r < 1, \frac{1}{k+1} < \theta < \frac{1}{k} \right\}.$$

Define

$$I_k := \{f \in \mathcal{A}_n \mid \forall j > k, f|_{S_j} = 0\}$$

Clearly this is an ideal:

- Let f, g vanish on all $z \in S_j$ for all $j > k$. Then $f(z) + g(z) = 0 + 0 = 0$, so $f + g \in I_k$.
- Let $f \in I_k$ and $g \in \mathcal{A}_n$. Then for any $z \in S_j$ with $j > k$, we have $f(z)g(z) = 0g(z) = 0$. Thus $f \cdot g \in I_k$.

Note that if $f \in I_k$, then it vanishes on S_j for all $j > k$. In particular, it vanishes for $j > k + 1$, so $f \in I_{k+1}$. Thus $I_k \subseteq I_{k+1}$ for all k . We show this inclusion is proper.

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be any nonzero smooth bump function supported in $(0, 1)$. We define $\psi_k : \mathbb{C}^n \rightarrow \mathbb{C}$ by

$$\psi(z_1, \dots, z_n) = \exp\left(-\frac{1}{|z_1|}\right) \cdot h(\arctan(\operatorname{Im} z_1 / \operatorname{Re} z_1)),$$

and $\psi(0, z_2, \dots, z_n) = 0$. Since the exponential goes to 0 as $z_1 \rightarrow 0$, its derivatives will also go to zero, and thus the function is smooth. Note that the arctangent term yields the angle θ for the polar representation of the first coordinate. We can then take h to be supported on some subset of $(1/(k+2), 1/(k+1)) \subseteq (0, 1)$, meaning ψ_k will vanish on all S_j with $j > k + 1$ since its support is disjoint from the S_j s. Thus $\psi_k \in I_{k+1}$. Note that the support of ψ_k can only intersect $\bigcup_{j=1}^k \overline{S_j}$ at 0 due to the disjointness of the sets $(1/(j+1), 1/j)$. Note also that by construction, ψ_k doesn't vanish inside of S_{k+1} . Since $0 \notin S_k$ but is in its closure, there is no neighborhood which we can restrict to make ψ_k vanish on S_{k+1} . Thus $\psi_k \notin I_k$, and the inclusion is proper. Thus $\mathcal{A}_n$ does not satisfy the ascending chain condition, and is not Noetherian. ■

Problem 5 (1.13). *In contrast to the case of holomorphic mappings between domains in $\mathbb{C}$, it is not the case that in general a holomorphic map between domains in higher dimensions is an open mapping. For example, show that*

$$f : \mathbb{C}^2 \rightarrow \mathbb{C}^2, \quad (z, w) \mapsto (z, zw)$$

is not an open mapping.

Proof. Let $B_\varepsilon(0) \subseteq \mathbb{C}^2$ be an open ball about 0 of radius $\varepsilon > 0$. We claim that $f(B_\varepsilon(0))$ is not open in $\mathbb{C}^2$. Recall that a basis for a topology has the property that given any open set U and any $x \in U$, there is an element of the basis $x \in B \subseteq U$. Note also that $\mathbb{C}^2 \cong \mathbb{R}^4$ has a basis given by open balls. We show this is not true for $0 \in f(B_\varepsilon(0))$, proving that the set is not open.

For any $\delta > 0$ and $B_\delta(0) \subseteq \mathbb{C}^2$, there exists $(0, \frac{\delta}{2}) \in B_\delta(0)$. Note also that if $f(z, w)$ is of the form $(0, zw)$, then $z = 0$ and thus $f(z, w) = 0$. Thus there exists no $(0, w) \in f(B_\varepsilon(0))$ with $w > 0$, meaning $(0, \frac{\delta}{2}) \notin f(B_\varepsilon(0))$. Thus $B_\delta(0)$ is not a subset of $f(B_\varepsilon(0))$. ■